AutoCallable Analytics Platform

Pricing & Risk Analysis for Structured Products

Monte Carlo | Finite Difference | Heston Calibration | Dupire Local Vol

Rohit Padebettu · June 2026

Demo App

autocallable-pricer
.streamlit.app

Deng et al. (2011)

Haugh (2013)

Alm et al. (2013)

AUTOCALLABLE

What is an Autocallable?

The core idea

An autocallable is a structured note tied to an index (e.g. the S&P 500). At each observation date, if the index is above a barrier, the note redeems early and pays a coupon. If not, it continues.

Three possible outcomes at maturity:

- Early call — index above barrier on any observation date. Investor receives notional + coupon.
- Capital protection — index below barrier but above the knock-in floor. Investor gets notional back, no coupon.
- Knock-in loss — index breached the protection floor. Investor shares in the index's downside.

Why do investors buy them? Autocallables offer significantly higher coupons than bonds in exchange for taking on downside exposure to equities.

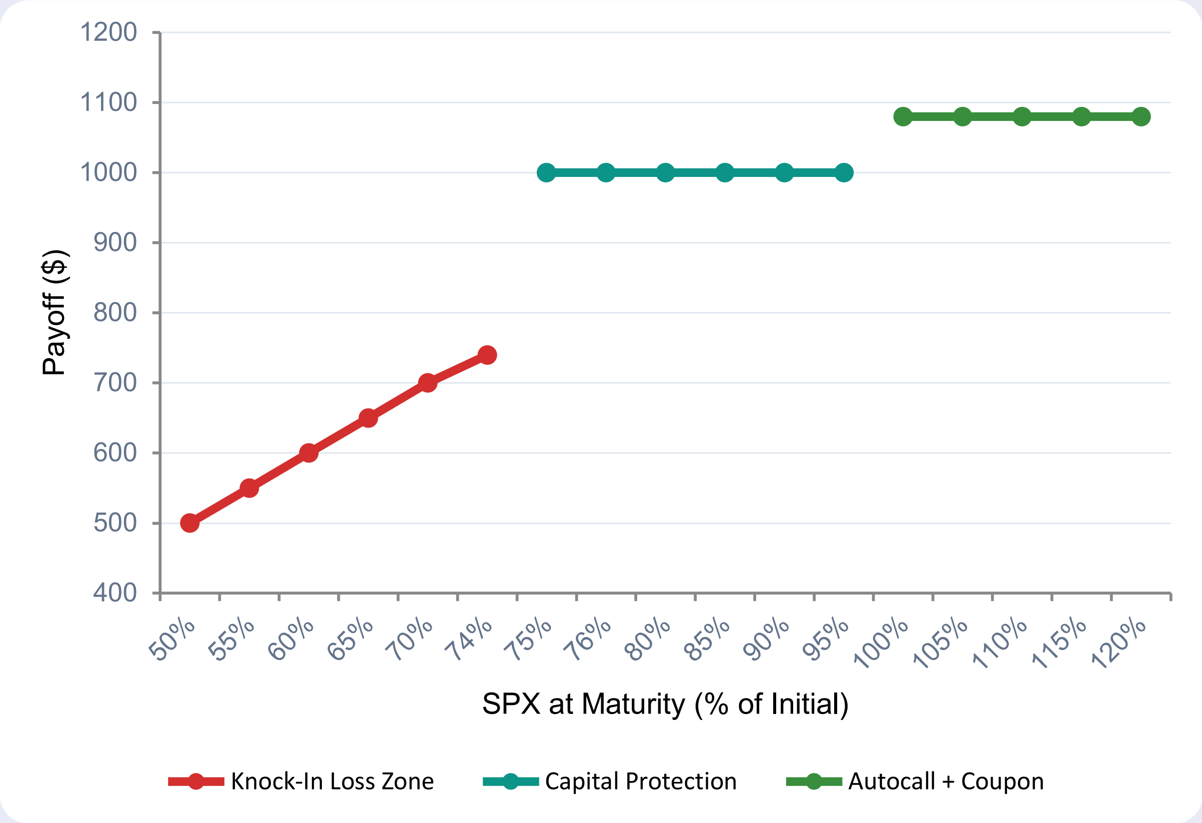
Phoenix Autocall — Term Sheet

Used throughout this presentation

Term	Value
Underlying	S&P 500 (SPX)
Notional	$1,000
Maturity	2 Years
Observation	Quarterly (8 dates)
Call Barrier	100% of initial
Coupon	8% p.a. (2% per quarter)
Protection	75% Knock-In Put
Mkt Snapshot	Jun 6 2026 (SPX ≈ 7,400)

PAYOFF STRUCTURE

Phoenix Autocall — Payoff at Maturity



Below 75%

Knock-In Loss

Investor receives SPX return — full downside. A $1,000 note is worth $500 if SPX falls 50%.

75% – 100%

Capital Protected

SPX fell but stayed above the 75% floor. Investor gets $1,000 back. No coupon, but no loss.

Above 100%

Autocall Triggered

SPX at or above barrier on a quarterly date. Investor receives $1,000 + 2% coupon per quarter elapsed.

Settings: S_0 = 5,600 · Call Barrier = 100% · Knock-In = 75% · Coupon = 8% p.a. (payoff shown for year-2 maturity call)

VOLATILITY

Four Volatility Models — Why It Matters

The Black-Scholes model assumes constant volatility — but real market option prices imply different vol at every strike and maturity. Autocallable barriers sit at specific strikes: using the wrong vol misprices the product.

Flat Volatility

Baseline

Black-Scholes

One constant vol for all strikes and maturities. Fast, simple — but provably wrong. Sets the baseline for comparison.

Local Volatility

Haugh (2013) Eq. 2

Dupire (1994)

A different vol at each (price, time) point — derived from real market option prices. Matches every traded price by construction.

Heston Stoch. Vol

Haugh (2013) Eq. 23

Heston (1993)

Volatility is random: it has a mean level, a reversion speed, its own randomness (vol-of-vol), and correlation to the stock. Captures equity skew naturally.

Bates (Heston + Jumps)

Haugh (2013) §4

Heston + Merton

Adds occasional crash 'jumps' to Heston's stochastic vol. Fits the short-dated implied smile that Heston alone can struggle with.

VOL SURFACE

Volatility Surface — What the App Shows

Tab 1

3D Implied Vol Surface

Real SPX options data filtered and plotted as an interactive 3D surface — moneyness × maturity × implied vol. Rotate it to see the equity skew in both dimensions. This is smoothened using cubic spline

Tab 2

Heston Model Overlay

After 'Calibrate Heston,' the 5-parameter Heston surface is overlaid on the market surface. Red patches show fit error. Calibration uses L-BFGS-B (~15–45 sec).

Tab 3

Dupire Local Vol Surface

Derived from the implied vol surface via Dupire's formula. Looks flatter in the strike dimension — it accounts for future path dynamics, not just today's prices. The paper recommends to use SVI before differentiating, but the app uses cubic splines, to illustrate the noise

Tab 4

Model Comparison

Calibrate Heston, Merton, and Bates in sequence. RMSE table compares fit quality. Smile charts show which model fits each maturity best.

Snapshot: SPX Options · Jun 6 2026 22:38 (Market Close) · ~1,400 option quotes · Moneyness 0.80–1.20 · Maturities 1m–18m

PRICING: FDM

Pricing Method 1 — Finite Difference (FDM)

How it works

Think of it as a spreadsheet filled in from right to left.

We know the exact payoff at maturity. The FDM starts there and works backwards through time — at each step computing what the note is worth one step earlier, given possible stock moves.

When spot crosses the call barrier at an observation date, the boundary condition fires automatically — the note is marked called.

Two schemes: Explicit FD (simple) and Crank-Nicolson (more stable — blends current and next step).

Ref: Deng, Mallett, McCann (2011) §2.2

Call Probability Term Structure

P(call at this date | not called earlier)

Date	Period	Prob.
Q1	3 months	34.2%
Q2	6 months	13.8%
Q3	9 months	7.9%
Q4	12 months	5.1%

Q5–Q7: 3.6% · 2.6% · 1.9% · Cumulative ≈ 69%. ~31% reach maturity.

App: FDM Visualization page — watch V(S, t) evolve backwards via heatmap and time-slice animation. Scheme Comparison tab benchmarks Explicit vs Crank-Nicolson: convergence as grid refines, stability, and timing per pricing call.

Settings: $\sigma = 18\%$ · $r = 4.3\%$ · $q = 1.5\%$ · $S_0 = 5{,}600$ · $T = 2$ yr · Grid: 1,000 spatial × 500 time steps

PRICING: MC

Pricing Method 2 — Monte Carlo Simulation

Standard Monte Carlo

Simulate thousands of SPX paths. For each path, check every observation date — if above the barrier, the note calls and pays the coupon. Average the discounted payoffs.

The catch:
Accuracy scales as $1/\sqrt{N}$. At 1K paths: ±\$4. At 100K paths: ±\$0.40. Every 100× more paths buys only 10× more precision.

Variance reduction: Antithetic variates pair each random path with its mirror, roughly halving the variance for free.

Ref: Alm, Harrach, Harrach, Keller (2013) Eq. 2.3

One-Step Survival MC

Instead of randomly crossing the barrier, each path analytically computes the probability of crossing it at every step — and prices the autocall event without ever simulating a hit.

The advantages:
3–5× tighter confidence bands at the same path count — because the payoff is a smooth, continuous function of the starting price.

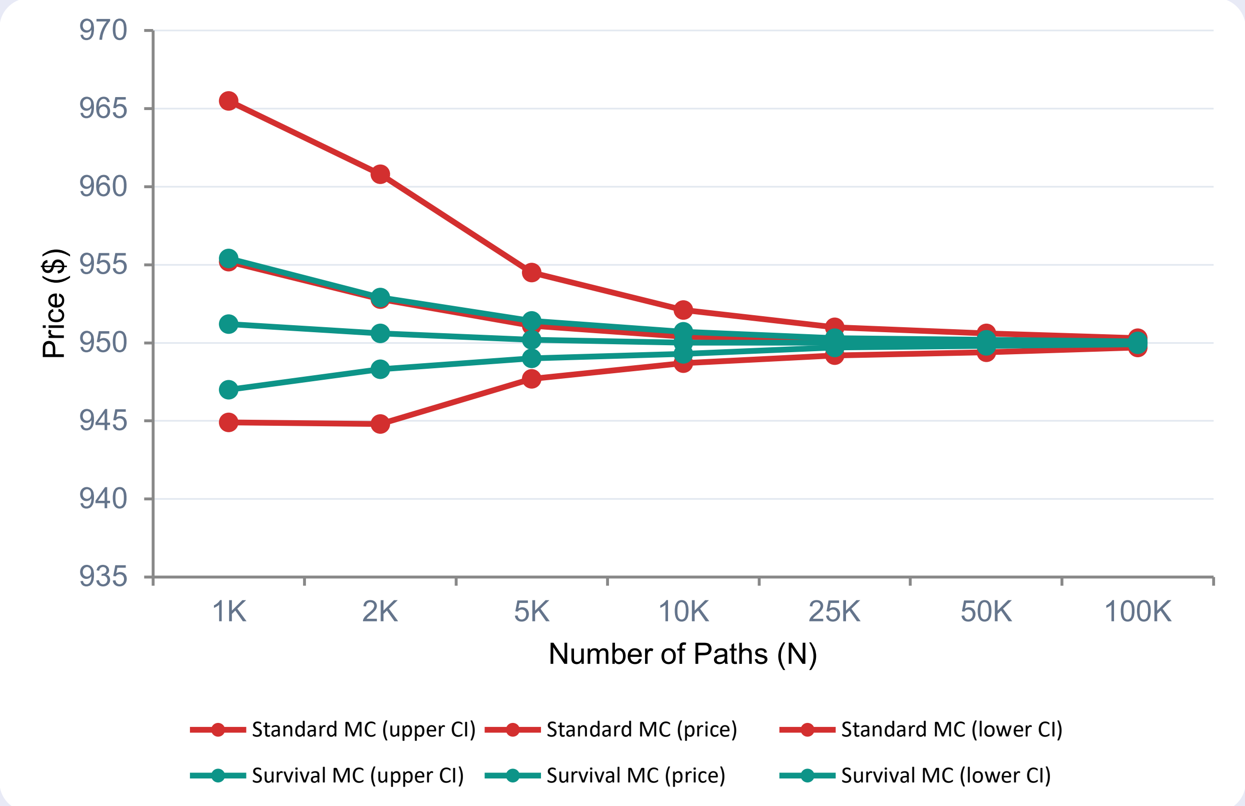
Stable Greeks: Delta and Vega are smooth and reliable — not noisy and sign-flipping as in standard MC. Usable for hedging.

Ref: Alm, Harrach, Harrach, Keller (2013) Algorithm 1

Settings: Phoenix Autocall · $\sigma = 18\%$ · $r = 4.3\%$ · $q = 1.5\%$ · $S_0 = 5{,}600$ · $N = 10{,}000$ paths · Antithetic: On

VARIANCE REDUCTION

Variance Reduction — Survival vs Standard MC



What the chart shows

- **Standard MC** — wide band at low N. Needs ~50K+ paths to stabilize.
- **Survival MC** — band is 3–5× narrower at every N. Same accuracy with far fewer paths.
- **Both converge** to the same price (~$950). Difference is precision, not correctness.

At N = 10,000 paths:
Std MC: ±$1.68 · Survival: ±$0.62

Settings: Phoenix Autocall · Flat vol σ = 18% · r = 4.3% · q = 1.5% · S_0 = 5,600 · Antithetic: On · Seed: 42

RESULTS

Pricing Results — All Methods, All Models

Pricing Method	Flat Vol (σ=18%)	Dupire Local Vol	Heston Stoch. Vol	Bates (Heston + Jumps)
FDM / PDE	$962	$951	$962 ‡	$962 ‡
Standard MC	$958 ±$4.2	$944 ±$2.1	$949 ±$1.7	$936 ±$2.4
Survival MC	$957 ±$1.1	$938 ±$0.8	$927 ±$0.6	$921 ±$1.0

‡ FDM uses flat vol as proxy for Heston/Bates (full stochastic-vol PDE requires 2D S×v grid — future extension). ± = 95% confidence interval.

Model choice matters

Flat vol vs Bates can differ by $40+ on a $1,000 note — a 4% pricing gap. This is real model risk, not noise. The vol model is the biggest driver after the term sheet itself.

Variance reduction works

Survival MC consistently beats Standard MC on precision at the same N. The ±$0.62 vs ±$1.68 intervals at 10K paths reflect 3–5× lower variance — by construction, not by luck.

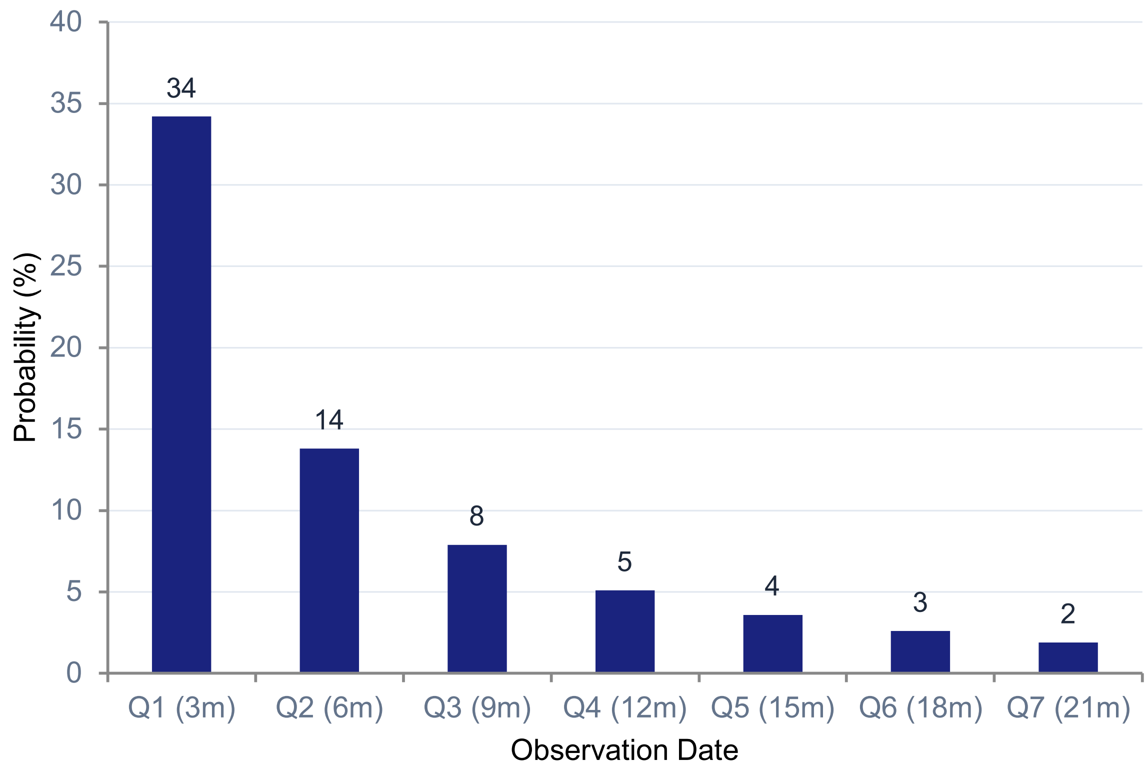
Cross-validation confirms

With the same vol model (flat), FDM, Standard MC, and Survival MC all converge to ~$958. This confirms correctness — no single method is trusted blindly.

Settings: Phoenix Autocall · S_0 = 5,600 · r = 4.3% · q = 1.5% · N = 10,000 paths · T = 2 yr · Snapshot: Jun 6 2026

CALL PROBABILITIES

Call Probability by Observation Date



Reading this chart

Each bar is the probability of the note calling at exactly that date — conditional on not having called earlier.

Key takeaways:

- ~34% chance of calling at Q1 alone — the single most likely outcome
- ~69% cumulative early-call probability across all 7 dates
- ~31% chance of reaching maturity — where knock-in risk applies

This term structure is the key output of the FDM analytical approach. It drives the product's effective duration and coupon yield.

Settings: FDM analytical · σ = 18% · r = 4.3% · q = 1.5% · Call Barrier = 100% · S_0 = 5,600

ADDITIONAL FEATURES

Additional Features — Explore in the App

Greek Stability — Pg. 4

The problem

Standard MC Delta near a barrier is unstable — tiny changes in bump size can flip the sign. Unusable for hedging.

The fix

Survival MC eliminates barrier crossings analytically. The payoff is smooth in S_0, so Delta and Vega are clean and monotone at every bump size.

Try it

Side-by-side plots at 10 bump sizes × 8 random seeds. Standard MC is spiky. Survival MC is smooth every time.

FDM Visualization (Page 3)

Value heatmap

V(S,t) plotted as a 2D heatmap — stock on x-axis, time on y-axis, color = value. The autocall barrier appears as a sharp discontinuity edge.

Time-slice animation

A slider sweeps the value function backwards from maturity (T) to today — you watch the algorithm work in real time.

Scheme comparison

Explicit FD vs Crank-Nicolson: price convergence as grid refines, CFL stability behavior, and per-call timing benchmarks.

Custom Product Builder (Page 6)

Design your own

Choose structure type (Phoenix, Step-Down, Digital), maturity, barriers, coupon, and observation frequency. All in one form.

Live payoff preview

Payoff diagram updates instantly as you change any parameter — no Run button needed.

Price it

Save to the sidebar. The custom product immediately appears in the Pricer page and can be priced with all three methods.

Architecture & References

Research Foundation

1. **Deng, Mallett, McCann (2011)**

   PDE for autocallables. FDM on log-transformed heat equation. Closed-form continuous autocall. Call probability term structure.

2. **Haugh, Columbia IEOR (2013)**

   Heston characteristic function (Eq. 23 — branch-cut safe). Dupire local vol. Merton jump-diffusion. Bates model.

3. **Alm, Harrach, Harrach, Keller (2013)**

   One-step survival MC. GHK importance sampling. Variance reduction. Stable Greeks via barrier-free payoff construction.

Tech Stack

UI	Streamlit
Numerics	NumPy · SciPy · Pandas
Calibration	scipy.optimize L-BFGS-B
Charts	Plotly (interactive)
Data	Pre-collected SPX snapshots
Tests	pytest — 74 tests passing
Deploy	Streamlit Community Cloud

Live Demo https://autocallable-pricer-nk7zy5ayx9amw77jrnxkau.streamlit.app

Start on the Pricer page · Select Phoenix Autocall · Hit Run All Pricers · Explore Vol Surface for Heston calibration